How to Kernelize SVM

Example: Quadratic Kernel

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \; X_i^T X_j$$

$$w = \sum_{i=1}^{N} \alpha_i y_i X_i \qquad b = -\frac{1}{2}\left[\max_{i:\, y_i = -1} W^T X_i + \min_{i:\, y_i = +1} W^T X_i \right]$$

Class of vector $Z \rightarrow \text{sign}(W^T Z + b)$

Kernelize using

$K(a, b) = (a^T b)$

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{i=1}^{N} \sum_{j=1}^{N} \alpha_i \alpha_j y_i y_j \; [X_i^T X_j]^2$$

$$w = \sum_{i=1}^{N} \alpha_i y_i X_i \qquad b = -\frac{1}{2}\left[\max_{i:\, y_i = -1} [W_i^T X_j]^2 + \min_{i:\, y_i = +1} [W_i^T X_j]^2 \right]$$

Class of vector Z is $\text{sign}([W^T Z] + b)$